

2425-MA140 Engineering Calculus

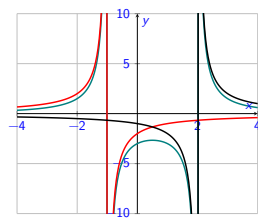
Week 2, Lecture 1

Partial Fractions

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This version of the slides are by Niall Madden, but are adapted from original notes by Dr Kirsten Pfeiffer.

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News!

Tutorials

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Partial Fractions

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Exercises

For more, see Section 2.5 (Rational Functions) of *Modern Engineering Mathematics*:
https://search.library.nuigalway.ie/permalink/f/3b1kce/TN_cdi_askewsholts_vlebooks_9780273742517

- Tutorials start **this** week. The schedule is:
- Teams 1+2: Tuesday 15:00 ENG-2003
 - Teams 3+4: Tuesday 15:00 ENG-2034
 - Teams 9+10: Thursday 11:00 ENG-2002
 - Teams 11+12: Thursday 11:00 ENG-3035
 - Teams 5+6: Friday 13:00 Eng-2002
 - Teams 7+8: Friday 13:00 Eng-2035

If you are interested to taking a tutorial through Irish, please complete this survey: <http://tinyurl.com/suirbhe1>

- There is currently a “practice” assignment open. See <https://universityofgalway.instructure.com/courses/35693/assignments/94873>
- During tutorials, the tutor will solve some similar questions. You can access the tutorial sheet at https://universityofgalway.instructure.com/courses/35693/files/2023552?module_item_id=650912
- A new assignment will open by tomorrow...

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In addition, each week I'll post a set of exercises related to the material covered. You don't have to submit your work for these, but you should try them: they are similar in style and standard to exam questions.

Rational Functions have the general form $f(x) = \frac{p(x)}{q(x)}$, where $p(x)$ and $q(x)$ are polynomials.

An (proper) rational function can often be written as a sum of simpler ones: **partial fractions**.

For example

$$\frac{8x - 12}{x^2 - 2x - 3}$$
$$\frac{3}{x - 3} + \frac{5}{x + 1}$$

can be written as

In order to do this, we try to **factorize** the denominator.

Note: Any polynomial (with real coefficients) can be factorised fully into the product of

- linear
- and irreducible quadratic factors.

We get different combinations of factors in the denominator. Let's look at **four cases**, and how to find the partial fractions in each case.

Partial Fractions

The four cases

1. Linear factors to the power of 1 in the denominator.
2. Linear factors to the power greater than 1 in the denominator, (i.e repeated linear factors).
3. Irreducible quadratic factors.
4. Irreducible quadratic factors to power greater than 1.

1: Eg $f(x) = \frac{p(x)}{q(x)}$ $q(x) = (x-2)(x-4)$.

2. $q(x) = (x+2)(x^2+2x+1) = (x+2)(x+1)^2$

3. $q(x) = x^2 + 1$

4. $q(x) = (x^2+1)^2$

Partial Fractions

Case 1

(1) Linear factors to the power of 1 in the denominator.

Example

$$\frac{3x}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$$

Find A and B.

Method 1: Compare Coef (ie of $1=x^0$ and of x) by matching powers of x . Multiply the terms on the right, above + below, to get $(x-1)(x+2)$ in the denominator:

$$\frac{3x}{(x-1)(x+2)} = \frac{A(x+2)}{(x-1)(x+2)} + \frac{B(x-1)}{(x-1)(x+2)}$$

Partial Fractions

Case 1

There are **two methods** for finding A and B.

Method 1: Comparing coefficients (continued).

$$\begin{aligned} \frac{3x+0}{(x-1)(x+2)} &= \frac{A(x+2)}{(x-1)(x+2)} + \frac{B(x-1)}{(x-1)(x+2)} \\ &= \frac{Ax + 2A + Bx - B}{(x-1)(x+2)} \\ &= \frac{(A+B)x + (2A-B)}{(x-1)(x+2)} \end{aligned}$$

$$\begin{aligned} \text{So } A+B &= 3 \\ 2A-B &= 0 \\ \hline 3A &= 3 \\ A &= 1, B = 2 \end{aligned}$$

Partial Fractions

Case 1

Method 2: Substituting specific values for x .

Recall $\frac{3x}{(x-1)(x+2)} = \frac{A(x+2)}{(x-1)(x+2)} + \frac{B(x-1)}{(x-1)(x+2)}$.

So $3x = A(x+2) + B(x-1)$ for all x .

(a) Pick $x=1$. $\Rightarrow 3(1) = A(1+2) \Rightarrow 3=3A$
 $A=1$

(b) Pick $x=-2$. $\Rightarrow 3(-2) = A(0) + B(-2-1)$
 $\Rightarrow -6 = -3B$
 $B=2$

So $\frac{3x}{(x-1)(x+2)} = \frac{1}{x-1} + \frac{2}{x+2}$.

Partial Fractions

Case 1

Example

Write $\frac{8x-12}{x^2-2x-3}$ as sum of partial fractions.

Step 1: factorise x^2-2x-3 as $(x-3)(x+1)$.

$$\frac{8x-12}{x^2-2x-3} = \frac{A}{x-3} + \frac{B}{x+1} = \frac{A(x+1) + B(x-3)}{(x-3)(x+1)}$$

So $8x-12 = A(x+1) + B(x-3)$

(a) $x=3$ $8(3)-12 = A(4) \Rightarrow 12 = 4A \Rightarrow A=3$

(b) $x=-1$ $8(-1)-12 = B(-4) \Rightarrow -20 = -4B \Rightarrow B=5$
 Ans $\frac{8x-12}{x^2-2x-3} = \frac{3}{x-3} + \frac{5}{x+1}$

Partial Fractions

Case 1

Exercise 2.1

Find the constants A, B and C, so that

$$\frac{2x+1}{(x-2)(x+1)(x-3)} = \frac{A}{x-2} + \frac{B}{x+1} + \frac{C}{x-3}$$

Week 2, Exer 1.

Partial Fractions

Case 2

(2) Linear factors to the power greater than 1 in the denominator, (i.e repeated linear factors).

If $(x - \alpha)^k$ appears in the denominator, it will give rise to the following terms:

$$\frac{A_1}{x - \alpha} + \frac{A_2}{(x - \alpha)^2} + \dots + \frac{A_k}{(x - \alpha)^k}$$

Partial Fractions

Case 2

Example

Find A , B and C such that

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+2}$$

(Note: we'll find that $A = 5/9$, $B = 4/3$ and $C = -5/9$).

$$\frac{3x+1}{(x-1)^2(x+2)} = \frac{A(x-1)(x+2)}{(x-1)^2(x+2)} + \frac{B(x+2)}{(x-1)^2(x+2)} + \frac{C(x-1)^2}{(x+2)(x-1)^2}$$

So we see

$$3x+1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

Partial Fractions

Case 2

$$3x+1 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

$$\textcircled{a} \quad x=1 \quad 3(1)+1 = A(0)(3) + B(3) + C(0)^2$$

$$\Rightarrow 4 = 3B \quad B = 4/3$$

$$\textcircled{b} \quad x=-2 \quad \Rightarrow \quad 3(-2)+1 = A(-3)(0) + B(0) + C(-3)^2$$

$$\Rightarrow -5 = 9C \quad C = -5/9$$

So now we have

$$3x+1 = A(x^2+x-2) + (4/3)(x+2) + (-5/9)(x-1)^2$$

Now we match the powers of x and get $x^2(A - 5/9) = 0 \Rightarrow A = 5/9$

Partial Fractions

Case 3

(3) Irreducible quadratic factors.

Irreducible quadratic factors can not be factorised using real numbers, e.g. $x^2 + x + 1$.

An irreducible quadratic factor $ax^2 + bx + c$ gives rise to partial fractions of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

Example 2.34 from textbook

If one writes

$$\text{irreducible} \cdot \frac{5x}{(x^2+x+1)(x-2)} = \frac{Ax+B}{x^2+x+1} + \frac{C}{x-2}$$

then we find $A = 10/7$, $B = 5/7$ and $C = 10/7$.

Partial Fractions

Case 4

(4) Irreducible quadratic factors to power greater than 1.

Each repeated irreducible quadratic factor $(ax^2 + bx + c)^k$ in the denominator will give rise to

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \dots + \frac{A_kx + B_k}{(ax^2 + bx + c)^k}$$

These can be done in a similar way to the previous case. But the calculations are pretty messy, so we won't even try!

Partial Fractions

Exercises

Exercise 2.2

Express the following as partial fractions.

- $\frac{6}{x^2 - x - 2}$
- $\frac{2x-1}{x^2 - x - 2}$
- $\frac{x-1}{(x+1)(x^2 - x - 2)}$
- $\frac{x}{x^2 + 2x + 1}$
- $\frac{1}{x^3 - 1}$

Finished here!